

SUPPLIER QUALITY INSPECTION

Good manufacturing practices inspection report

For-cause supplier quality inspection

INSPECTED ORGANIZATION BluePeak Biologics, Inc.	SITE Harbor Point Biologics Campus
INSPECTION DATES 11 January 2027 to 13 January 2027	REPORT ISSUED 22 January 2027
LEAD INSPECTOR Priya Nanduri, Director, Global Supplier Quality	SITE IDENTIFIER BPB-US-NH-01
TRIGGERED BY DEV-BP-1026-SUPPLIER · BP-1026-03 · TC-4	

Caldova confidential – supplier quality

FOR CAUSE

FINAL

Report status.

This is a fictional Caldova supplier-quality inspection report created for an AI retrieval demonstration. It is not issued by WHO or any medicines regulator and makes no regulatory certification claim.

PART 1 · GENERAL INFORMATION

Site, inspection basis, and scope

SITE ADDRESS 410 Harbor Point Parkway, Portsmouth, New Hampshire 03801, United States	EMPLOYEES 486
OPERATIONS Mammalian cell-culture drug-substance manufacturing, harvest, intermediate hold, and bulk drug-substance storage	
PRODUCTS AND CAMPAIGNS IN SCOPE Caldova campaign BP-1026; runs BP-1026-01, BP-1026-02, and BP-1026-03	
PREVIOUS CALDOVA AUDIT Routine Caldova qualification audit, 14–16 April 2025; two minor observations closed 30 June 2025	INSPECTION BASIS Quality agreement BPB-CAL-QA-2025 and supplier oversight procedure CAL-SQP-014

OBJECTIVE

Assess the site's investigation of the BP-1026-03 contamination event, determine whether the implicated transfer-path maintenance controls meet GMP and quality-agreement expectations, and evaluate whether containment and proposed corrective actions adequately protect Caldova product.

SCOPE

- Bioreactor suite BR-04 and associated transfer path to hold tank HT-04
- Transfer connection TC-4, hose-set maintenance, assembly, and pressure-hold verification
- Environmental and in-process microbiological monitoring for campaign BP-1026
- Deviation, CAPA, training, change-control, and maintenance records relevant to the event
- Supplier quality governance for contracted Caldova manufacturing

INSPECTION TEAM

- Priya Nanduri — Lead inspector, Global Supplier Quality
- Evan Cole — Principal Engineer, Biologics Manufacturing
- Marisol Vega — Senior Manager, Quality Systems

SITE REPRESENTATIVES

- Hannah Wells — Site Quality Head
- Marcus Lin — Director of Manufacturing
- Devon Price — Maintenance and Reliability Manager

Limitations. The inspection did not include unrelated suites, financial systems, clinical operations, or a full reinspection of all site GMP systems. Conclusions apply to the sampled systems and records.

PART 1 · EVENT AND EVIDENCE

Contamination event under inspection

BP-1026-03 Affected run	TC-4 Implicated connection	Ralstonia pickettii Organism
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EVENT DATE 19 December 2026	DETECTED BY In-process bioburden sample, vessel day 17
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IMMEDIATE ACTION Run isolated; transfer path and hold tank placed on quality hold.
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INVESTIGATION ROOT CAUSE A supplier-maintained transfer-hose connection was reassembled with a pinched gasket after preventive maintenance, compromising the closed path.

EVIDENCE CONFIRMED DURING INSPECTION

- Positive sample localized downstream of transfer connection TC-4
- Retained upstream sample remained negative
- Gasket inspection found a visible compression fold
- Maintenance closeout omitted the required post-assembly pressure-hold result

Commercial cross-reference. \$111,000 contamination-investigation charge on invoice line INV-SUP-009-2026-Q4-L004; commercial disposition is handled separately

RECORDS REVIEWED

- Batch production records BP-1026-01 through BP-1026-03
 - Deviation DEV-BP-1026-SUPPLIER and supporting microbiology data
 - Work order WO-BPB-1026-441 and pressure-hold form PHT-BPB-1026-441
 - Transfer hose assembly SOP BPB-MNT-117, revision 6
- Closed-system integrity SOP BPB-PRD-204, revision 4
 - Training matrices for BR-04 manufacturing and maintenance personnel
 - Environmental monitoring trends for BR-04, October–December 2026
 - Change controls and CAPA effectiveness checks sampled from 2026

LINKED IDENTIFIERS

INVOICE	INV-SUP-009-2026-Q4	INVOICE LINE	INV-SUP-009-2026-Q4-L004
PURCHASE ORDER	PO-SUP-009-2026-Q4	BATCH	BATCH-BP-1026
AFFECTED RUN	BP-1026-03	SUITE	BR-04
DEVIATION	DEV-BP-1026-SUPPLIER	DEVIATION PACKET	DEV-BPB-1026
UTILIZATION LOG	UTL-BPB-1026	CAPACITY	CAP-BP-2026-Q4
CONNECTION	TC-4	HOLD TANK	HT-04
MAINTENANCE ORDER	WO-BPB-1026-441	PRESSURE TEST	PHT-BPB-1026-441
CAPA	CAPA-BPB-2027-003		

PART 2 · INSPECTION OBSERVATIONS

GMP systems assessment

SYSTEM	INSPECTION ASSESSMENT	RATING
Pharmaceutical quality system	The site initiated deviation controls promptly and notified Caldova within the quality-agreement timeline. Investigation governance was weakened by late Quality involvement in maintenance evidence review and by an initial root-cause statement issued before the connection was disassembled.	Improvement required
Personnel and training	Operators interviewed understood line clearance and aseptic response expectations. Two of six maintenance technicians had not completed the revision-6 practical assessment for sanitary gasket installation before performing independent work.	Deficient
Premises and equipment	BR-04 was orderly and appropriately segregated. TC-4 is positioned below an overhead return bend, limiting direct visual confirmation of gasket seating after assembly. No inspection mirror or borescope step was specified.	Deficient
Production and in-process controls	Manufacturing records showed expected upstream controls and timely isolation. The closed transfer path was released to service without documented evidence that the required pressure-hold acceptance criterion had been met.	Deficient
Laboratory control	Microbiology methods, controls, and sample traceability reviewed were acceptable. Localization of the positive result downstream of TC-4 and the negative retained upstream sample were scientifically coherent with the identified breach.	Acceptable
Documentation and data governance	The maintenance system permitted closure of WO-BPB-1026-441 while the linked pressure-hold result field was blank. The audit trail recorded the omission but did not trigger supervisory escalation.	Deficient
Materials, sanitation, and hygiene	Gasket receipt, storage, status labeling, cleaning, and personnel gowning controls sampled were acceptable. No evidence indicated incoming gasket material failure or deficient room sanitation.	Acceptable
Qualification, validation, and maintenance	The transfer path had an approved qualification basis, but maintenance verification was treated as a record-completion activity rather than a mandatory equipment-release gate. The control did not prevent use after incomplete testing.	Deficient
Complaints, recalls, contracts, and self-inspection	No released Caldova material was implicated and no recall was initiated. Contract-notification controls operated as designed. The 2026 self-inspection sampled maintenance timeliness but did not test evidence completeness or release interlocks.	Improvement required

Assessment convention. “Acceptable” indicates no material deficiency in the sampled evidence. “Improvement required” identifies a weakness captured in an Other deficiency or recommendation. “Deficient” maps to a formal deficiency on the following pages.

PART 3 · DEFICIENCIES

Major deficiencies

BPB-MAJ-01 · Closed transfer path released without required integrity-test evidence

MAJOR

The site released transfer path TC-4 for manufacturing use after maintenance order WO-BPB-1026-441 although pressure-hold record PHT-BPB-1026-441 contained no result, reviewer confirmation, or objective evidence that the acceptance criterion was met.

Objective evidence: Blank pressure-hold result; work-order closure audit trail; run BP-1026-03 equipment-release entry.

Response due: 05 February 2027

Potential risk: Loss of closed-system integrity and microbial contamination of in-process product.

Required response: Root cause, immediate correction, systemic corrective action, accountable owner, due date, and proposed effectiveness check.

BPB-MAJ-02 · Maintenance qualification did not ensure reliable sanitary-gasket installation

MAJOR

The practical qualification program did not prevent technicians without current revision-6 assessment from independently assembling product-contact connections, and the procedure lacked a compensating visual-verification method for TC-4's restricted orientation.

Objective evidence: Training matrix sample; technician interviews; direct observation of TC-4 access; compressed gasket recovered from the connection.

Response due: 12 February 2027

Potential risk: Recurring assembly defects may remain undetected until process monitoring identifies contamination.

Required response: Root cause, immediate correction, systemic corrective action, accountable owner, due date, and proposed effectiveness check.

Classification basis. A Major deficiency is a non-critical departure that has produced, or may produce, product not consistently meeting requirements, or a combination of departures indicating a significant systems failure. No Critical deficiency was identified in the inspected scope.

PART 3 · DEFICIENCIES

Other deficiencies and conclusion

BPB-OTH-01 · Quality review entered too late in the maintenance evidence workflow

OTHER

Quality review occurred during deviation investigation rather than before post-maintenance equipment release for this high-risk product-contact connection.

Objective evidence: Workflow configuration and role-permission review.

Response due: 19 February 2027

Potential risk: Incomplete evidence may not be challenged before manufacturing use.

BPB-OTH-02 · Self-inspection sampling did not test maintenance-record completeness

OTHER

The 2026 self-inspection checked overdue work orders but did not sample mandatory linked test results or failed release conditions.

Objective evidence: Self-inspection plan SIA-BPB-2026 and checklist review.

Response due: 05 March 2027

Potential risk: Systemic record-completeness weaknesses may remain undetected.

0 Critical	2 Major	2 Other
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OVERALL CONCLUSION

GMP compliance cannot be considered fully satisfactory for the inspected transfer-path controls until major deficiencies are corrected and effectiveness is verified.

Supply status. Conditional approval: no new Caldova BR-04 run may begin until interim controls for gasket verification, witnessed pressure-hold testing, and Quality release are documented and accepted by Caldova.

PART 3 · RESPONSE AND CLOSURE

Corrective and preventive action commitments

COMMITTED ACTION	OWNER	DUE	EFFECTIVENESS EVIDENCE
Revise equipment-release workflow so required pressure-hold results and Quality approval are hard-stop fields.	Hannah Wells	05 Feb 2027	Challenge test in validated maintenance system using a deliberately incomplete record
Replace TC-4 gasket and hose set; execute witnessed pressure-hold test before return to service.	Devon Price	27 Jan 2027	Approved protocol, raw pressure data, and Quality release
Retrain and practically reassess all technicians who assemble sanitary product-contact connections.	Devon Price	12 Feb 2027	Direct observation using seeded gasket-position defects
Add mirror or borescope confirmation and second-person verification for restricted-orientation connections.	Marcus Lin	12 Feb 2027	Three consecutive compliant assemblies at TC-4
Expand self-inspection sampling to linked evidence completeness and release-interlock effectiveness.	Hannah Wells	05 Mar 2027	Independent audit of 20 post-maintenance equipment releases

Closure condition. Submission of a plan does not close a deficiency. Caldova Global Supplier Quality will verify implementation and effectiveness before changing conditional supply status.

ATTACHMENTS

ID	TITLE	SOURCE
Attachment A	BR-04 process flow and contamination-risk pathway	Inspector reconstruction from batch record, piping walkdown, sample locations, and deviation evidence
Attachment B	Inspection photograph — TC-4 gasket seating	Synthetic demonstration image generated for this fictional report
Attachment C	Inspection photograph — pressure-hold verification station	Synthetic demonstration image generated for this fictional report

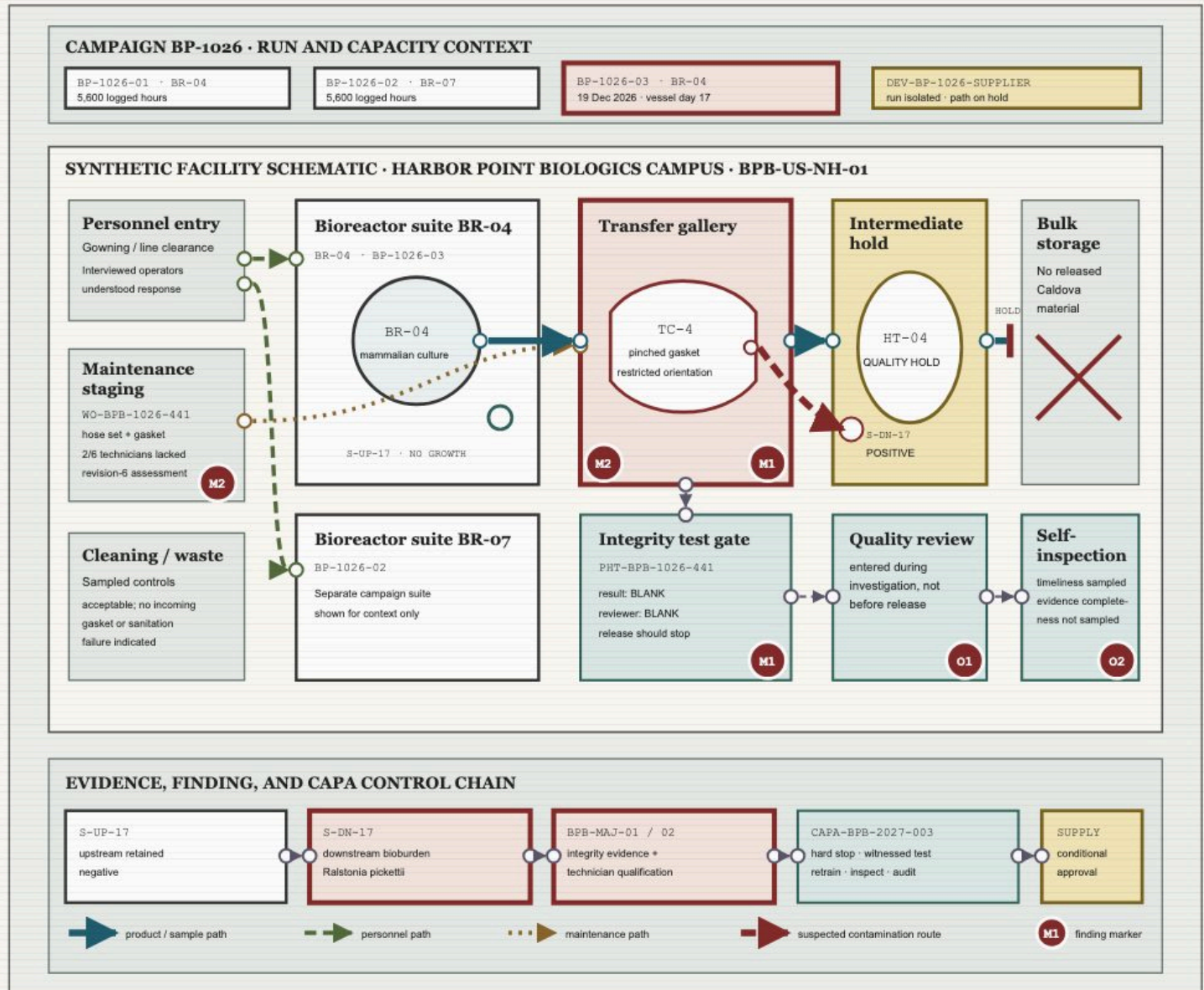
Priya Nanduri
Lead inspector, Caldova Global Supplier Quality
Date: 22 January 2027

Hannah Wells
Site Quality Head, BluePeak Biologics; receipt acknowledged
Date: 22 January 2027

ATTACHMENT A · FIGURE 1

BR-04 facility, control, and contamination-risk map

Inspector reconstruction from batch records, maintenance evidence, microbiology results, interviews, and piping walkdown. Spatial arrangement and support-room adjacency are synthetic, schematic, and not to scale.



Evidence interpretation. The negative upstream retained sample, positive downstream sample, and recovered gasket defect support TC-4 as the most probable entry point. The map distinguishes observed process evidence from the synthetic room arrangement.

Control interpretation. Missing pressure-hold evidence and late Quality review allowed the maintenance path to bypass a release gate. CAPA adds a hard stop, witnessed testing, practical reassessment, visual verification, and completeness sampling.

ATTACHMENT B · PHOTOGRAPH 1

TC-4 gasket seating condition



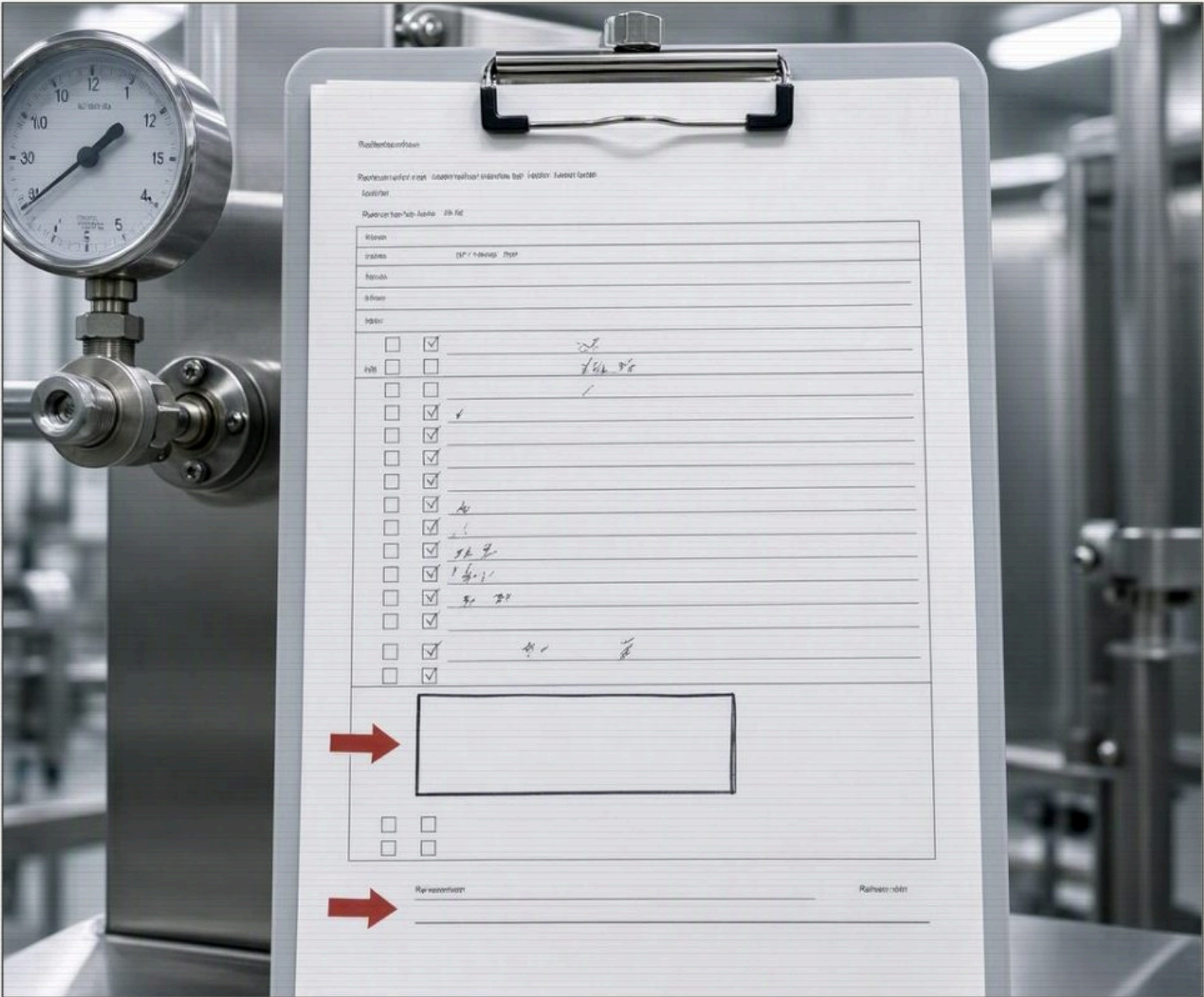
Evidence ID: IMG-BPB-TC4-01	Related deficiency: BPB-MAJ-02
Location: BR-04 transfer connection TC-4	Orientation: Product-contact sanitary connection

Observed condition. The image illustrates uneven elastomer exposure around the sanitary joint, consistent with a gasket compressed out of its intended seat. Restricted access at the installed orientation made direct visual confirmation difficult without an inspection aid.

IMAGE DISCLOSURE: This is an original synthetic demonstration image generated for the fictional report. It is not a photograph of an actual BluePeak facility, product, person, or event.

ATTACHMENT C · PHOTOGRAPH 2

Pressure-hold verification station



Evidence ID: IMG-BPB-PHT-01	Related deficiency: BPB-MAJ-01
Record: PHT-BPB-1026-441	Equipment release: WO-BPB-1026-441

Control context. The checklist appears substantially completed, but the test-result box and reviewer-confirmation line remain empty. The pressure gauge and test connection provide the means to verify closed-path integrity; without recorded results and review evidence, the work order should not have permitted equipment release.

IMAGE DISCLOSURE: This is an original synthetic demonstration image generated for the fictional report. It is not a photograph of an actual BluePeak facility, product, person, or event.